

Research paper

Optimal transient power sharing method for shipboard MVDC System based on segmented variable step dynamic programming

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ARTICLE INFO

Keywords:

MVDC
Segmented variable step
Dynamic programming
Transient power sharing

ABSTRACT

Supplying power for large pulsed power load with shipboard medium voltage direct current integrated power system has continued to be a challenge. The energy storage device configured in the system can effectively mitigate the power impact on grid and gas turbine gensets, but it also complicates the system operation, requiring advanced power sharing and control strategies for support. In this paper, the control framework of system configured with hybrid energy storage is presented, the objective function to quantify transient power disturbance on gas turbine genset is established, and the segmented variable step dynamic programming method for system transient power sharing is innovatively proposed and employed to derive the optimal transient power-sharing scheme for the rectangular and triangular large pulsed power load conditions. Compared with the conventional approach through MATLAB/Simulink simulation, the scheme derived from segmented variable step dynamic programming demonstrates enhanced utilization of the transient power compensation capability of hybrid energy storage, and leads to a reduction in the power regulation pressure on gas turbine gensets. In the assumption of two typical operating processes, as the results of the power impacts, the transient regulation rate of the genset rotor is reduced from 10.4 % to 9.9 % and from 13.7 % to 8.6 %, respectively, the fatigue damage of shaft system is decreased by 53 % and 94.4 %, respectively, and the maximum inlet relative temperature of gas generator turbine is reduced by 0.015 and 0.139, respectively.

1. Introduction

The shipboard medium voltage direct current integrated power system (Simplified MVDC system) centrally governs and distributes the ship's power, provides better conditions for the use of shipboard high energy weapons, and represents the advanced direction of the world naval ship power platform development (Khazaei, 2021). With the growing demand for the practical application of various types of shipboard large pulsed power load (LPPL), energy storage has evolved into an indispensable and pivotal system component, as well as imposing elevated demands on system power regulation and energy management (Duan et al., 2019).

LPPL is typically represented by electromagnetic rail gun, lasers and radar system, with the power used featuring pulsed characteristics, high peak and large ramp rate (Khan et al). In comparison, the main power source of MVDC system is gas turbine gensets with relatively slow power regulation capabilities. When LPPL-weapons operating or high-power loads switching, the ramp rate of grid load is up to 100 MW/s. The

generator will closely track the load power to maintain a stable bus voltage, leading to a direct transfer of power impact onto gas turbine and inducing severe shaft speed fluctuation. In accordance with the grid frequency transient tolerance specified in STANAG 1008 (Pearson and Newman, 2021) and Mil Spec 1399 (Roa), as well as the proportional correlation between alternating current (AC) frequency and prime mover speed, the transient speed regulation (TSR), denoting the maximum percentage of speed deviation during transients, cannot exceed $\pm 4\%$. Although direct current (DC) system accomplishes the decoupling between speed and generation frequency, TSR of the prime mover cannot exceed a certain limit, commonly $\pm 10\%$. If the rotational speed fluctuates beyond the tolerance, it will exert an influence on the operation of genset and the quality of electricity (Liu et al., 2024). Furthermore, swift and frequent changes in generator output power prompt rapid alterations in shaft torque and internal state of gas turbine, resulting in life degradation. It can be concluded that the gas turbine represents a significant vulnerability of the genset in terms of its independent response to LPPL.

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Received 6 May 2024; Received in revised form 2 July 2024; Accepted 11 July 2024

Available online 16 July 2024

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For LPPL operation, it is standard practice to configure the system with energy storage for compensating transient power, emphasizing the proper coordination of system power flows (Shi et al., 2018). However, most of the current research on energy management and coordinated control of hybrid power systems focuses on mitigating voltage fluctuations. (Zhuo, 2021) adopted a voltage-based active and passive charge/discharge switching strategy to manage the flywheel operation. (Stone et al) enhanced the system power quality during pulse load operation by reducing high frequency fluctuation of generator output power. (Alafnan et al., 2018) suggested a hybrid energy storage system (HESS) combination of superconducting magnet energy storage and battery, and designed a dynamic droop control strategy based on state of charge (SOC) to scheme the charging and discharging sequence, which improved the DC power stability. There are other researches mainly focusing on improving the operational efficiency. (Van Vu et al., 2017) effectively reduced the power loss by optimizing the power curves of the genset and energy storages. (Liu et al., 2023) proposed a system transient flow optimization method, which could effectively reduce configuration cost of energy storage. The adaptive model proposed in (Hou et al., 2018) aimed at reducing the wear of the motor drive shaft.

System energy management relies on appropriate controls, rooted in specific rules, for example, the operating state of the power sources are classified according to ship power, SOC or efficiency maps and controlled according to set rules (Hou et al., 2021). Further, (Khan et al., 2017) adopted fuzzy rule-based hybrid energy storage control strategies to improve the adaptability to cope with different operating conditions. Intelligent optimization algorithms prove effective in resolving nonlinear control challenges through iterative exploration of parameter space. For instance, (Yang et al., 2021) proposed a new control technique based on the hybridization of improved particle swarm optimization algorithm and cuckoo search algorithm to tune the fuzzy PID controller parameters. To improve the control adaptivity, (Zhu et al., 2020) put forward a modified adaptive equivalent consumption minimization strategy. Based on the model predictive control algorithm, (Hou et al., 2018; Hou et al., 2019) could effectively capture the parameter variations and further improve the efficiency and reliability of the system. To achieve global optimization, dynamic programming (DP) algorithm is widely used in system management to support power sharing decisions. (Zeng et al., 2018) considered the economic operation of microgrid and programmed the optimal schedule for a day. (Zhang and Xiong, 2015) implemented adaptive energy management for real-time driven period based on 100 s of historical information.

To conclude the current research, the main objectives of coordinated control and energy management for MVDC system focus on enhancing voltage transient stability and reducing costs. However, for LPPL scenarios, such considerations are insufficient since the main problem of transient impact on gas turbine is omitted. In terms of control strategies and optimization algorithms, the fact that LPPL operates with pre-known power curve, distinguishing it from other random power disturbances, results in DP algorithm may provide an advantage over the optimal control algorithm based on system state feedback. Still, the critical point that remains is that traditional DP primarily addresses economic objectives during long-term system operation. Its direct application to solving transient power sharing problems, especially considering transient objectives, presents challenges.

Based on the above analyzes, this paper proposes a transient power sharing method with the objective of mitigating the transient impact on gas turbine with limited energy storage configurations. There are two main contributions: firstly, the proposal of an appropriate system coordinated control architecture is suggested; Secondly, a segmented variable-step dynamic programming method is innovatively proposed to obtain the optimal transient power sharing scheme for LPPL operation, which results in a less transient impact to gas turbine as compared to conventional optimal control schemes.

2. Coordinated control and transient power sharing of MVDC system

During shipboard LPPL operations, the power consumption of the grid changes substantially and rapidly. The transient process needs to rely on cooperation between genset and energy storage to mitigate power impact on grid and gas turbine and to meet system transient stability requirements. In this section, the relevant theoretical methods of coordinated control and power sharing involved in this transient process are sorted out and summarized.

2.1. System topology and components

Fig. 2.1 depicts a notional topology of MVDC system, which mainly involves of a gas turbine genset, grid-connected HESS consisting of batteries and supercapacitors (SC), propulsion load, LPPL and service load (Khersonsky et al., 2010). The various grid connected components are integrated into the DC bus through converters to achieve an orderly power flow and coordinated operation controlled by a hierarchical control system. This structure has been simplified compared to the actual system in some details such as the number of gensets, but the energy coupling rules and operation control strategy can be adequately represented, so this paper mainly discusses and analyses based on this structure.

2.2. Control description

For MVDC system, the centralized control structure is suggested, as shown in Fig. 2.2, where V_{ref} is voltage reference value, V_{dc} is actual voltage value, n_{ref} is rotor speed reference value, n is actual rotational speed, P_{Bref} , P_{SCref} are power reference values of battery and supercapacitor respectively.

The genset is powered by a gas turbine, with power and voltage regulated by the excitation system and connected to the DC bus through an uncontrolled rectifier. The gas turbine adjusts the fuel based on the speed gap and tracks the generator output power. The power regulation of batteries and supercapacitors relies on PWM-controlled bidirectional DC/DC converters. When the LPPL is to be operated, the integrated control system concurrently sends power compensation commands to HESS in parallel with operation command to LPPL. This approach allows the HESS to participate more flexibly in genset transient power regulation, effectively mitigating the power impact. In addition, the integrated control system integrates control strategies for other operating conditions and functions, such as online regulation of power distribution and control of parallel operation.

2.3. Transient power sharing

As illustrated in Fig. 2.2, the power sharing strategy is carried by the integrated control system and implemented by sending control commands from the integrated control system to device controllers. As for transient power sharing in LPPL environment with limited energy storage configuration, the sharing strategy should mainly focus on transient impacts, exploit hybrid energy storage features and capabilities as much as possible, and aim at reducing transient indicators such as power ramp rate and TSR of genset.

In order to provide a further presentation of transient power sharing and to facilitate it as a comparison in the subsequent section, a traditionally commonly used frequency splitting (FS) filtering strategy (Guo et al., 2019) is initially introduced here. Due to inherent characteristics of HESS, the load power is decomposed by fluctuation frequency, of which the high-frequency part is responded by supercapacitors, the mid-frequency part is responded by batteries, and the remaining low-frequency part is responded by gas turbine genset. The system transient power sharing control based on FS is shown in Fig. 2.3.

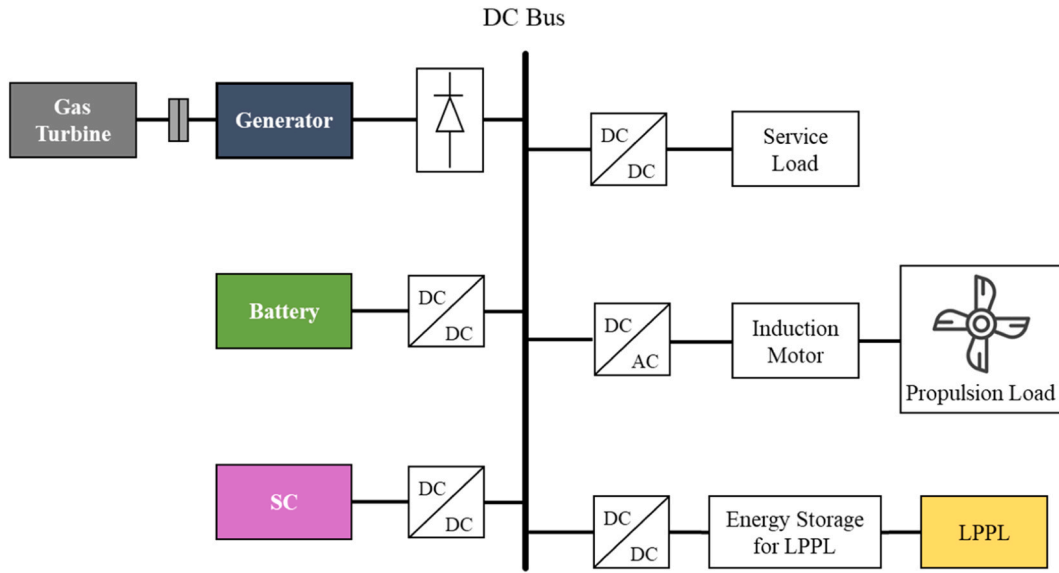


Fig. 2.1. Topology of the MVDC system.

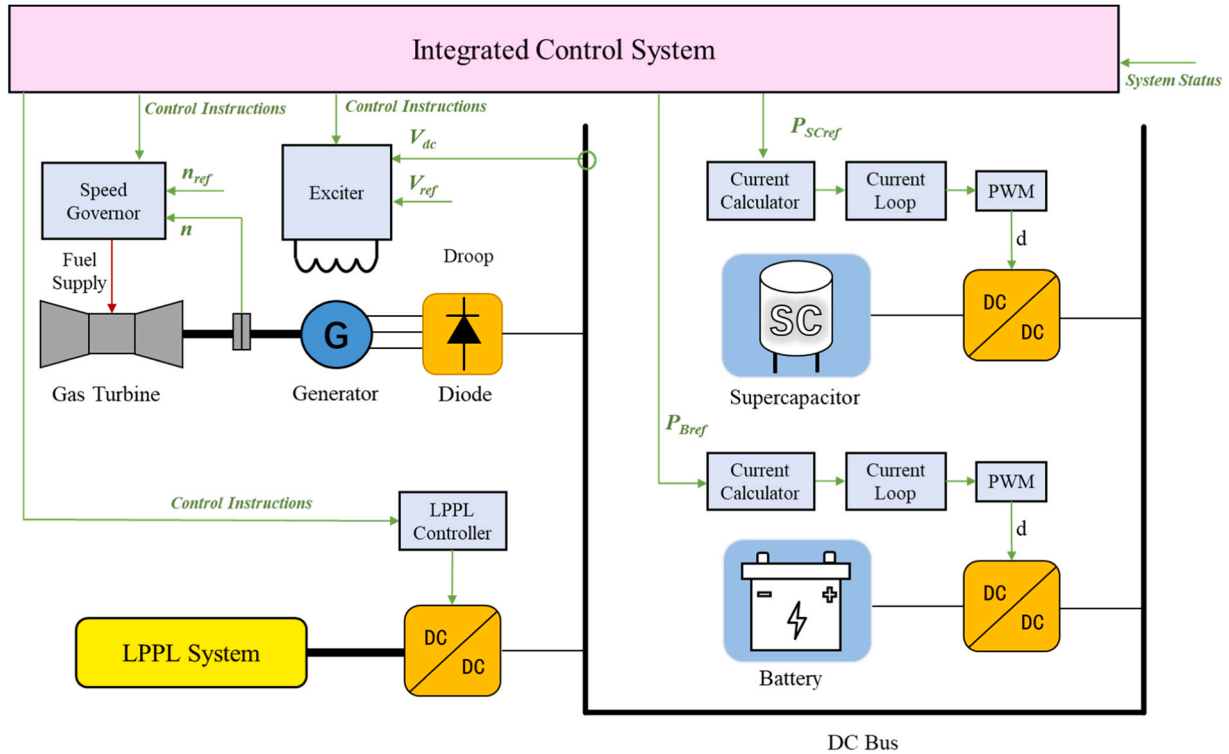


Fig. 2.2. Coordinated control structure for MVDC system.

$$P_{SCref} = K_{SC} \frac{s}{s + 1/T_{SC}} P_L \quad (2.1)$$

$$P_{Bref} = K_B \frac{s}{s + 1/T_B} (P_L - P_{SCref}) \quad (2.2)$$

$$P_G = P_L - P_{Bref} - P_{SCref} \quad (2.3)$$

It is known that programming the power sharing of three sources is to locate any two of the power curves. In Fig. 2.3, the central controller adopting the FS algorithm calculates the real-time power commands of the battery and supercapacitor, satisfying Eq. (2.1)(2.2), where P_L is the

load power which is obtained in real time from the closed-loop measurement. The genset desired power value P_G can be calculated according to Eq. (2.3) as an input to its power feedforward algorithm. The power sharing algorithm contains four main parameters, of which T_{SC} , T_B are time constants that determine the amount of power out from different sources. K_{SC} , K_B are power gain constants that would be used to limit the maximum charging and discharging power of energy storage. The rational power sharing curves can be obtained through optimal design of the above parameters, effectively dealing with various scenarios of load mutation in the MVDC system.

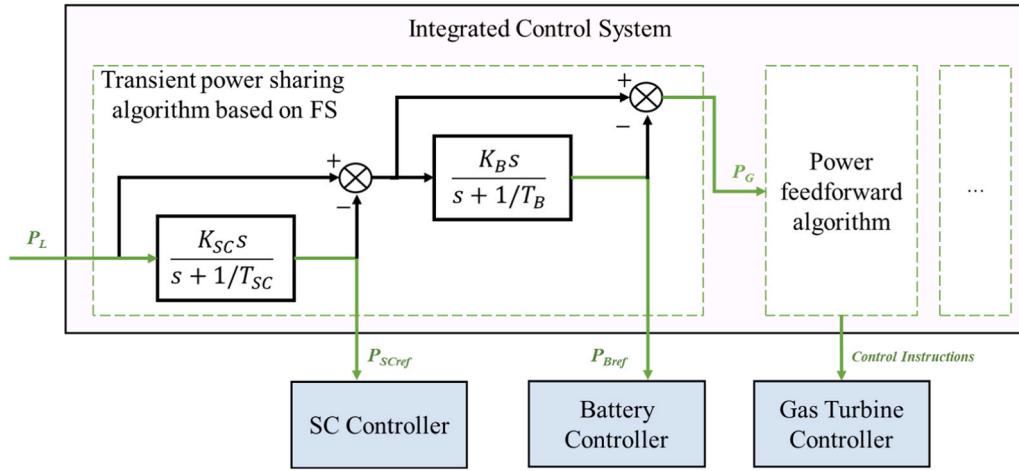


Fig. 2.3. Integrated control structure based on frequency splitting.

3. System transient power sharing optimization method based on SVSDP

The transient power sharing strategy based on FS method can flatten system power disturbances depending on HESS characteristics, contributing to impact mitigation on grid and genset. However, FS is a passive feedback power regulation method with high reliability, but poorly devise optimal scheme for specific operating conditions, such as LPPL, the power of which is pre-known. DP is well adapted to solving global optimization problems. However, in the application of DP to energy management, power is generally used as a decision variables and SOC as state variables. The objective function is a function of the state variables. Therefore, the optimal curve solution often assumes to disregard the power change process of the source, treating each moment of the optimization period, such as hours or days, as a steady state. This can pose challenges in directly applying DP to solve transient power sharing problems. In this section, a transient power sharing method based on segmented variable step dynamic programming (SVSDP) is proposed, offering an approach for the application of DP in transient objectives solving.

The system transient power sharing control based on SVSDP is shown in Fig. 3.1. The central controller sends the real-time power commands of the battery and supercapacitor, satisfying Eqs. (3.1)–(3.3).

$$P_{SCref} = f_{SC}^{P_L}(t^*) \quad (3.1)$$

$$P_{Bref} = f_B^{P_L}(t^*) \quad (3.2)$$

$$P_G = P_L - P_{Bref} - P_{SCref} \quad (3.3)$$

Where $f_{SC}^{P_L}(t)$, $f_B^{P_L}(t)$ are the power function of time for supercapacitor and battery in response to P_L , respectively, t^* is the time generated from zero timing following receipt of the trigger signal. As in Eq. (3.1)(3.2), the SVSDP control strategy is based on LPPL triggering. The key is to obtain the optimal power sharing scheme based on the pre-known power curve of LPPL, that is, to obtain $f_{SC}^{P_L}(t)$ and $f_B^{P_L}(t)$, which is presented in detail in the following.

3.1. DP for transient power sharing

Dynamic Programming, based on the Bellman principle, decomposes the global optimal problem-solving into several single-stage optimal decisions. The fundamental recursive equation is as follows:

$$J(x(k)) = \min_{u(k)} \{ C_{node_obj}(u(k), x(k)) + J(x(k+1)) \} \quad (3.4)$$

Where k represents stage number, $x(k)$ represents state variable of system in stage k , $u(k)$ represents the decision variable in stage k . $J(x(k))$ is the stage optimal objective function, representing the optimal objective value in state $x(k)$. $C_{node_obj}(u(k), x(k))$ is the stage cost function, representing instantaneous objective value when transitioning from state $x(k)$

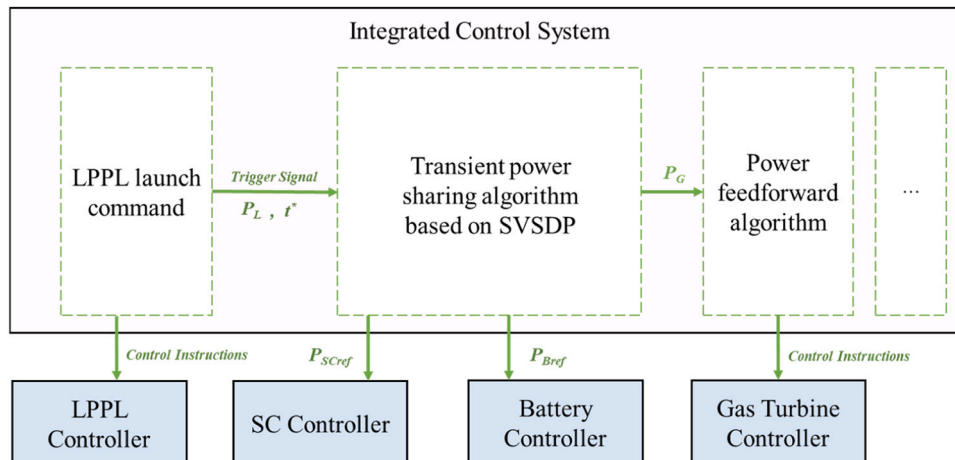


Fig. 3.1. Integrated control structure based on SVSDP.

Table 3.1
Power battery cell parameters.

Parameters	Values
Rated voltage (V)	2.3
Rated capacity (Ah)	10
Internal resistance (mΩ)	0.2
Charging rate	60 C
Maximum discharging current (A)	800
Weight (kg)	1
Size (mm ³)	60×608×118

to state $x(k+1)$ with $u(k)$.

To apply DP to transient power sharing in MVDC system, it is crucial to design an objective function that accurately captures the transient process. The objective can be described as maximizing the transient power compensation utility of the HESS, with the goal of minimizing power disturbances in genset during LPPL operation. In this paper, the metric presented in (Crider and Sudhoff, 2010) is referred. The genset power value at the last stage is innovatively introduced into DP state variables, which enables to quantify genset power disturbances in objective function, as in Eq. (3.5).

$$J = \min \sum_{k=1}^N F(X(k), U(k), P_L(k)) = \min \sum_{k=1}^N \left[\frac{P_G(k) - P_G(k-1)}{P_{Gref}} \right]^2 / \Delta T \quad (3.5)$$

Accordingly, the system state variables $X(k)$ is designed as in Eq. (3.6) and the decision variables $U(k)$ as in Eq. (3.7).

$$X(k) = \begin{bmatrix} X_1(k) \\ X_2(k) \\ X_3(k) \end{bmatrix} = \begin{bmatrix} P_G(k-1) \\ SOC_B(k) \\ SOC_{SC}(k) \end{bmatrix} \quad (3.6)$$

$$U(k) = \begin{bmatrix} U_1(k) \\ U_2(k) \end{bmatrix} = \begin{bmatrix} P_B(k) \\ P_{SC}(k) \end{bmatrix} \quad (3.7)$$

Where $P_L(k)$ is the load power in stage k , $P_G(k), P_B(k), P_{SC}(k)$ are output power of the genset, battery and supercapacitor respectively in stage k , P_{Gref} is power reference value of genset, $SOC_B(k), SOC_{SC}(k)$ are the SOC of battery and supercapacitor respectively, ΔT is the discrete step size.

$P_G(k)$ satisfies the power balance Eq. (3.8).

$$P_L(k) = P_G(k) + P_B(k) + P_{SC}(k) \quad (3.8)$$

The DP state transfer equation is shown in Eq. (3.9).

$$X(k+1) = \begin{bmatrix} P_G(k) \\ SOC_B(k+1) \\ SOC_{SC}(k+1) \end{bmatrix} = \begin{bmatrix} P_L(k) - P_B(k) - P_{SC}(k) \\ SOC_B(k) - I_B(P_B(k)) \cdot \Delta T / Q_B \\ SOC_{SC}(k) - I_{SC}(P_{SC}(k)) \cdot \Delta T / (U_{SC} C_{SC}) \end{bmatrix} \quad (3.9)$$

The battery and supercapacitor state transfer equation considers the SOC-based charging and discharging efficiency, and the relationship between the open-circuit voltage U_{oc} , output current I_o , output power P_o and internal resistance R_i of HESS satisfies Eq. (3.10), which can be calculated to obtain the efficiency equation as (3.11) (3.12).

$$I_o = \frac{U_{oc} - \sqrt{U_{oc}^2 - 4R_i P_o}}{2R_i} \quad (3.10)$$

$$\eta_{dis} = \frac{U_{oc} I_{dis} - I_{dis}^2 R_i}{U_{oc} I_{dis}} = \frac{1}{2} \left(1 + \sqrt{1 - \frac{4R_i P_o}{U_{oc}^2}} \right) \quad (3.11)$$

$$\eta_{chg} = \frac{U_{oc} I_{chg} - I_{chg}^2 R_i}{U_{oc} I_{chg}} = 2 \left/ \left(1 + \sqrt{1 - \frac{4R_i P_o}{U_{oc}^2}} \right) \right. \quad (3.12)$$

Table 3.2
Supercapacitor module parameters.

Parameters	Values
Rated voltage (V)	48
Rated capacity (F)	165
Internal resistance (mΩ)	2.3
Minimum operating voltage (V)	28.8
Energy density (Wh/kg)	3.9
Available power density (W/kg)	3800
Weight (kg)	13.5
Size (mm ³)	418×194×179

where η_{dis}, η_{chg} are the efficiencies of discharging and charging, I_{dis}, I_{chg} are the currents of discharging and charging.

Depending on the HESS characteristics, the system control variables are limited to the maximum and minimum values of the output power, and the state variables are also restricted to avoid overcharging and over-discharging of HESS. The relevant constraints are as follows.

$$\begin{cases} 0 \leq P_G(k) \leq P_{Gmax} \\ P_{Bmin} \leq P_B(k) \leq P_{Bmax} \\ P_{SCmin} \leq P_{SC}(k) \leq P_{SCmax} \\ SOC_{Bmin} \leq SOC_B(k) \leq SOC_{Bmax} \\ SOC_{SCmin} \leq SOC_{SC}(k) \leq SOC_{SCmax} \end{cases} \quad (3.13)$$

where $P_{Gmax}, P_{Bmax}, P_{SCmax}$ are the maximum output power of genset, battery and supercapacitor respectively, P_{Bmin}, P_{SCmin} are the minimum output power of battery and supercapacitor, SOC_{Bmin}, SOC_{Bmax} are the minimum and maximum SOC of battery, SOC_{SCmin}, SOC_{SCmax} are the minimum and maximum SOC of supercapacitor.

3.2. Improved DP design based on segmented variable step size

Applying DP to system transient energy management requires not only a transient objective function, but also with a reasonable discrete step size matching system power sources configuration and HESS's working period, as discussed in detail in Section 3.2.2.

3.2.1. Configuration of system power sources

In order to facilitate the following method description and results analysis, it is assumed that the system is configured with a gas turbine genset, storage of batteries and supercapacitors. The parameters of the battery cell are shown in Table 3.1 and the parameters of the supercapacitor module are shown in Table 3.2.

According to Table 3.1 and Table 3.2, the maximum charging power of single cell can reach 1.38 kW, and the maximum discharging power can reach 1.84 kW; the maximum charging and discharging power of supercapacitor module can be calculated as 51.3kw. According to Eq. (3.11)(3.12), the charging and discharging efficiencies of the battery

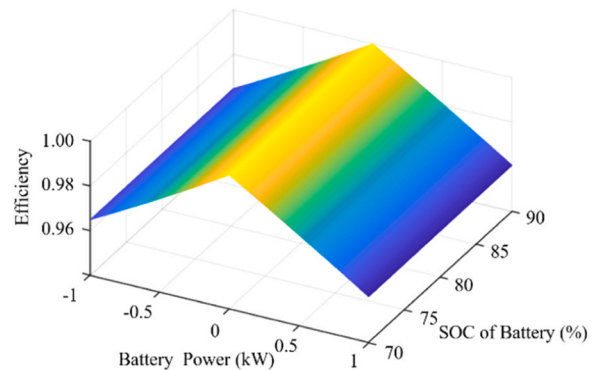


Fig. 3.2. Battery charging and discharging efficiency in relation to power and SOC.

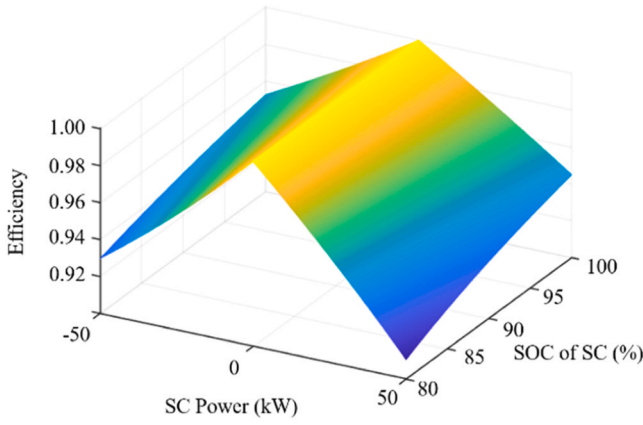


Fig. 3.3. Supercapacitor charging and discharging efficiency in relation to power and SOC.

and supercapacitor can be calculated as shown in Fig. 3.2 and Fig. 3.3.

The HESS is integrated with a number of battery cells and supercapacitor modules by series-parallel connection. For HESS health management and high efficiency operation, the maximum charging and discharging power and SOC of the batteries and supercapacitors are limited, as shown in Table 3.3, where SOC_{Bset} , SOC_{Bset} are the initial SOC of batteries and supercapacitors.

3.2.2. Segmented variable step design method

DP is solved using a numerical solution algorithm that requires discretization of the three-dimensional variables of running time, state variables and control variables. In general, large discrete intervals hurt solution accuracy, and small discrete intervals cause an exponential increase in calculations. Firstly, it's crucial to ensure consistency in power discretization intervals ($\Delta P_G, \Delta P_B, \Delta P_{SC}$) to prevent a situation where optimization results may be dominated by one of the larger intervals, as indicated in Eq. (3.14). Then, the discrete step size ΔT must be small enough to characterize the rate of load power mutations. Further, to characterize SOC changes, the power discretization intervals and the SOC discretization intervals $\Delta SOC_B, \Delta SOC_{SC}$ should satisfy Eq. (3.15) (3.16), where E_B, E_{SC} are the available energy of the batteries and supercapacitors.

$$\Delta P_G = \Delta P_B = \Delta P_{SC} \quad (3.14)$$

$$\Delta P_B \cdot \Delta T \geq E_B \cdot \Delta SOC_B \quad (3.15)$$

$$\Delta P_{SC} \cdot \Delta T \geq E_{SC} \cdot \Delta SOC_{SC} \quad (3.16)$$

It can be observed from Eqs. (3.14)–(3.16) that there is a dependence between discrete intervals of time, power and SOC. The discrete step size ΔT is mainly constrained by load curve and must be small enough. To avoid too small SOC discrete intervals, the discrete intervals of control variables $\Delta P_B, \Delta P_{SC}$ must be increased to reduce DP calculations. However, larger control variable discrete intervals need to be coupled with larger discrete step or the control variables will change excessively rapidly for an optimal solution to be derived. Here, the contradiction can be found generating. It can be concluded that the conventional discretization of DP can hardly cope with transient objective optimization problem, mainly due to constant ΔT cannot satisfy the multiple requirements from the load power, state variables and control variables. In response, a segmented variable step size improved design is proposed in this paper for optimal transient power sharing scheme, and the flow-chart is shown in Fig. 3.4. where N represents the total number of stages, $J(X(k, t), t)$ represents the stage optimal objective function selected, and $A(X(k, t), t)$ represents the corresponding control variables.

According to the method, the optimal power sharing schemes can be obtained for the pre-known load curves. It's worth noting that both the schemes and their correspondence with the loads are stored in the integrated control system, which sends down the corresponding instructions to execute the schemes when the loads are triggered to operate.

Then, the detailed process of the proposed method is introduced with a typical working condition. The relative load power curve is shown in Fig. 3.5, with a step from 0 to 1.0 during 0–0.1 s, and the programming period is 0–5 s.

According to power dynamic characteristics of the load, supercapacitors and batteries, it is appropriate to segment the DP period into three intervals, as shown in Fig. 3.6. The durations of the three intervals are T_0, T_1, T_2 , and the discrete steps are $\Delta T_0, \Delta T_1, \Delta T_2$, respectively. Meanwhile, Eqs. (3.5) and (3.9) have been updated to Eqs. (3.17) and (3.18), respectively, where $t(k)$ represents the corresponding time of stagek.

$$J = \min \left(\sum_{k=1}^{t(T_0)-1} \left[\frac{P_G(k) - P_G(k-1)}{P_{Gref}} \right]^2 / \Delta T_0 + \sum_{k=t(T_0)}^{t(T_0+T_1)-1} \left[\frac{P_G(k) - P_G(k-1)}{P_{Gref}} \right]^2 / \Delta T_1 \right. \quad (3.17)$$

$$\left. + \sum_{k=t(T_0+T_1)}^{N-1} \left[\frac{P_G(k) - P_G(k-1)}{P_{Gref}} \right]^2 / \Delta T_2 \right)$$

$$X \begin{pmatrix} k+1 \end{pmatrix} = \begin{cases} \begin{bmatrix} P_L(k) - P_B(k) - P_{SC}(k) \\ SOC_B(k) - I_B(P_B(k)) \cdot \Delta T_0 / Q_B \\ SOC_{SC}(k) - I_{SC}(P_{SC}(k)) \cdot \Delta T_0 / (U_{SC} C_{SC}) \end{bmatrix}, & 0 \leq t(k) < T_0 \\ \begin{bmatrix} P_L(k) - P_B(k) - P_{SC}(k) \\ SOC_B(k) - I_B(P_B(k)) \cdot \Delta T_1 / Q_B \\ SOC_{SC}(k) - I_{SC}(P_{SC}(k)) \cdot \Delta T_1 / (U_{SC} C_{SC}) \end{bmatrix}, & T_0 \leq t(k) < T_0 + T_1 \\ \begin{bmatrix} P_L(k) - P_B(k) - P_{SC}(k) \\ SOC_B(k) - I_B(P_B(k)) \cdot \Delta T_2 / Q_B \\ SOC_{SC}(k) - I_{SC}(P_{SC}(k)) \cdot \Delta T_2 / (U_{SC} C_{SC}) \end{bmatrix}, & T_0 + T_1 \leq t(k) \leq T_0 + T_1 + T_2 \end{cases} \quad (3.18)$$

Table 3.3
System power limits management.

Parameters	Values
Relative $P_{G \min}$	0
Relative $P_{G \max}$	2.0
Relative $P_{B \min}$	-0.2
Relative $P_{B \max}$	0.2
Relative $P_{SC \min}$	-0.6
Relative $P_{SC \max}$	0.6
SOC_{Bset} (%)	80
$SOC_{B \min}$ (%)	70
$SOC_{B \max}$ (%)	90
SOC_{SCset} (%)	90
$SOC_{SC \min}$ (%)	80
$SOC_{SC \max}$ (%)	100

The duration of each segment and the discrete step size should be defined subsequently. Firstly, according to $T_0 = \Delta T_0 \leq 0.1$, it is determined that: $T_0 = 0.1$ s, and then it can be calculated that $T_1 = 2 \times E_{SC} \cdot (SOC_{SCset} - SOC_{SC \min}) / P_{SC \max} = 1.2$ s, $T_2 = 5$ s - T_0 - $T_1 = 3.6$ s.

To define the remaining two key parameters $\Delta T_1, \Delta T_2$, the best group of them is explored in reasonable intervals. Table 3.4 demonstrates the DP results when different $\Delta T_1, \Delta T_2$ are selected. To exclude the effect of the remaining discrete parameters intervals, the discrete accuracies of control and state variables are taken to be: $\Delta P_G = 0.1, \Delta P_B = 0.1, \Delta P_{SC} = 0.1, \Delta SOC_B = 0.3\%, \Delta SOC_{SC} = 0.12\%$, the number of two decision variables is 10, the total number of each of the three state variables is 10, 331, 83 respectively.

From Table 3.4, it can be found that the values of $\Delta T_1, \Delta T_2$ affect the objective value and calculation time. The optimal group is $[\Delta T_1, \Delta T_2] = [0.2, 0.6]$, and the optimal transient power sharing scheme is shown in Fig. 3.7. (1) represents the DP result with constant $\Delta T = 0.1$ s, and the

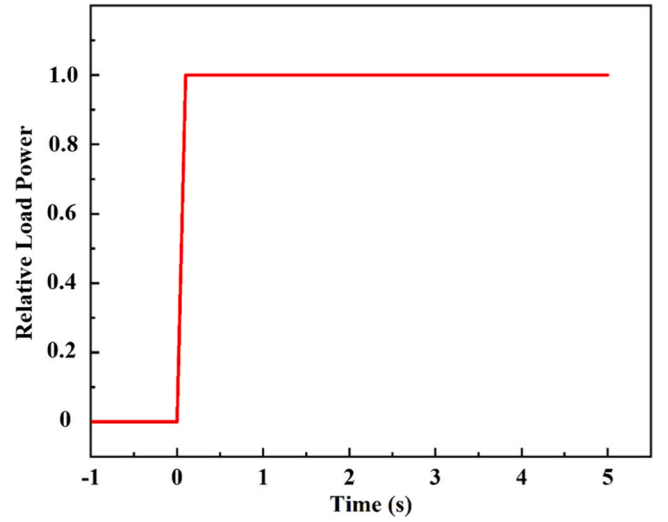


Fig. 3.5. Relative power curve for typical operating condition.

optimal transient power sharing scheme is shown in Fig. 3.8. Comparing (1) with (11), the superiority of the proposed segmented variable step-size design can be illustrated. With the transient power sharing scheme obtained by SVSDP, the genset power ramp rate is smaller and the calculation time is shorter. In order to avoid SOC errors caused by large discrete step size, the corresponding SOC of each group is reviewed, e.g., the group (18)(22)(23)(24), with better objective value, the corresponding parameters group cannot be adopted for SOC review failed.

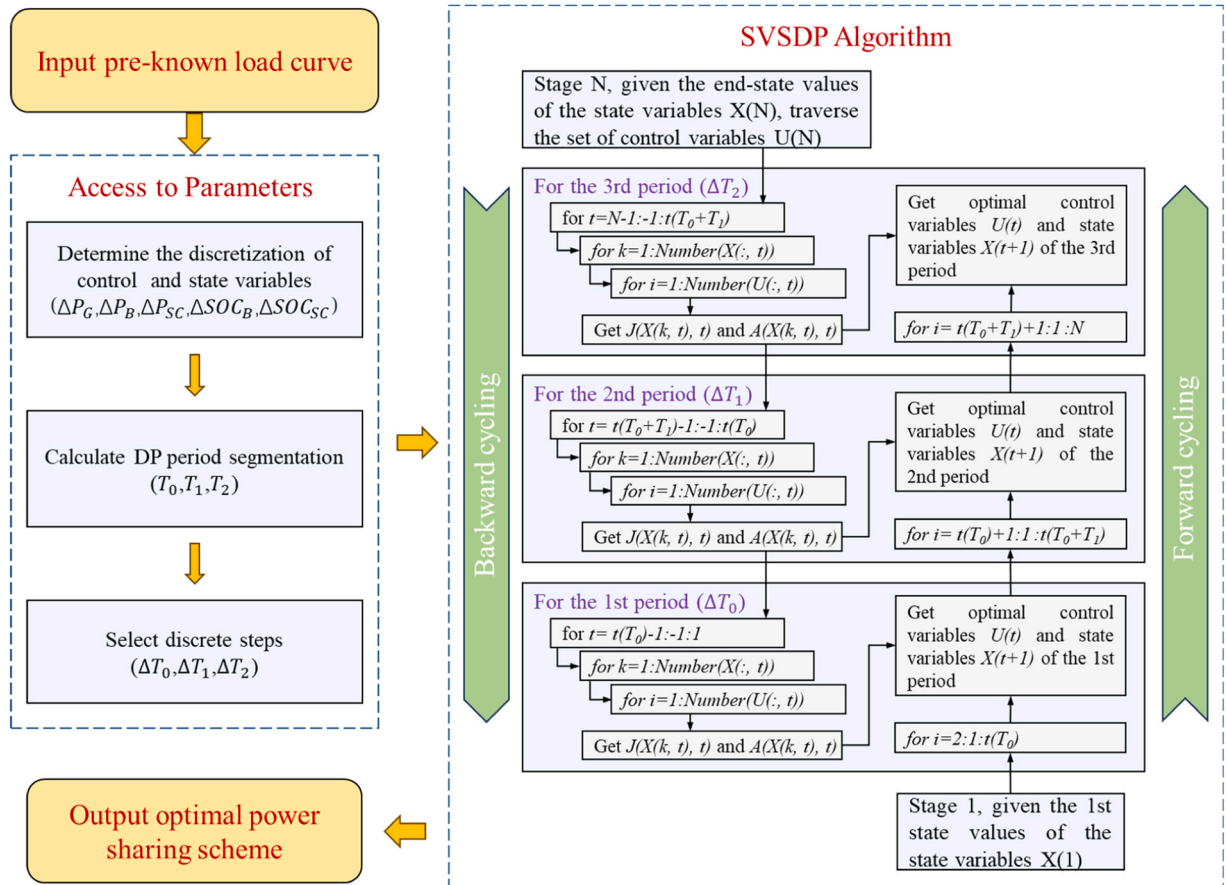


Fig. 3.4. Optimization method of transient power sharing scheme based on SVSDP.

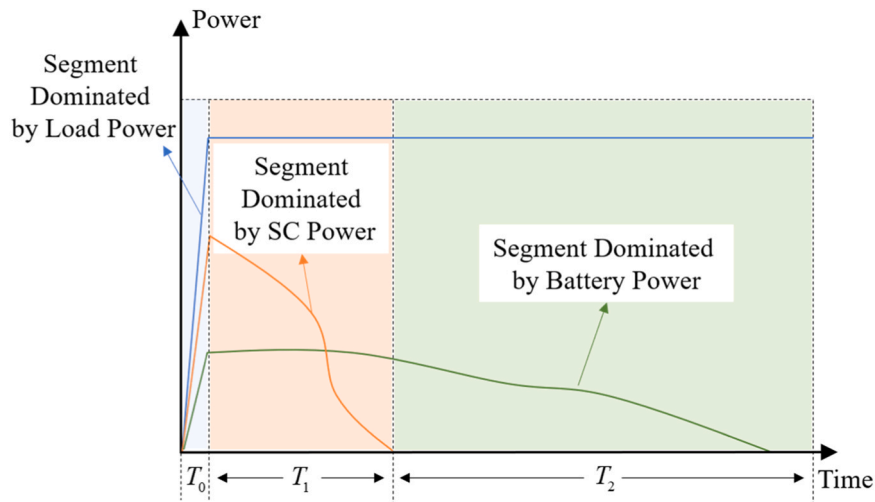


Fig. 3.6. Dynamic programming period segmentation scheme.

Table 3.4
DP results under different ΔT_1 , ΔT_2 .

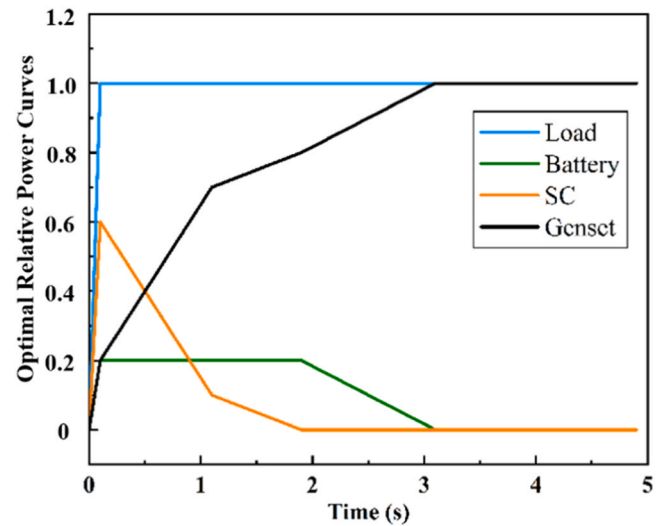
Parameters groups	ΔT_1 (s)	ΔT_2 (s)	Num of stage	Objective value	Calculation time (s)	SOC review
(1)	0.1	0.1	50	1.22	210	pass
(2)	0.1	0.2	32	1.10	123	pass
(3)	0.1	0.3	26	0.93	100	pass
(4)	0.1	0.4	23	0.97	87	pass
(5)	0.1	0.6	20	0.95	77	pass
(6)	0.1	1.2	17	0.94	64	failed
(7)	0.2	0.1	44	0.9	180	pass
(8)	0.2	0.2	26	0.8	97	pass
(9)	0.2	0.3	20	0.75	71	pass
(10)	0.2	0.4	17	0.73	62	pass
(11)	0.2	0.6	14	0.7	47	pass
(12)	0.2	1.2	11	0.69	36	failed
(13)	0.3	0.1	42	0.92	179	failed
(14)	0.3	0.2	24	0.93	93	failed
(15)	0.3	0.3	18	0.80	64	pass
(16)	0.3	0.4	15	0.77	49	pass
(17)	0.3	0.6	12	0.75	35	failed
(18)	0.3	1.2	9	0.67	26	failed
(19)	0.4	0.1	41	0.88	174	failed
(20)	0.4	0.2	23	0.77	89	failed
(21)	0.4	0.3	17	0.73	62	failed
(22)	0.4	0.4	14	0.7	45	failed
(23)	0.4	0.6	11	0.68	32	failed
(24)	0.4	1.2	8	0.67	20	failed

3.3. SVSDP comparative advantages analysis

In order to demonstrate the effectiveness and advantages of the proposed method, the transient power sharing scheme obtained by SVSDP and particle swarm optimization based frequency splitting (PSO-FS) are compared. The FS parameters are optimized by PSO and PSO basic parameters are shown in Table 3.5.

The PSO algorithm runs 5 times of the optimization process, and the global optimal fitness value of each group is ultimately settled around 0.78 as shown in Table 3.6, indicating the confidence of the optimization search result. The optimal individual fitness values and optimal parameters for the 5th group with the best result are shown in Fig. 3.9 and Fig. 3.10.

The objective value obtained by PSO-FS is 0.769, worse than the 0.70 of SVSDP scheme, which indicates that the system transient power sharing scheme obtained by SVSDP is superior than PSO-FS. Figs. 3.11–3.13 depict the parameter curves of the schemes based on SVSDP and PSO-FS.

Fig. 3.7. System relative transient power sharing curves with $\Delta T_1=0.2$ s, $\Delta T_2=0.6$ s.

In Fig. 3.11, during 0.2–2.5 s, the genset power based on SVSDP scheme is lower than that of PSO-FS scheme, causing less power impact. Figs. 3.12–3.13 reveal the explanation for that advantage. In comparison with the PSO-FS, the power ramp-up of SVSDP-based supercapacitors is more rapid during the 0–1 s period, exploiting the power-based advantage. Similarly, SVSDP-based batteries release more energy during transient period, making better use of the energy-based advantage.

4. Optimal system transient power sharing scheme in response to LPPL operation

In order to verify the effectiveness and improvement of the proposed method, for typical LPPL conditions, with system power configuration defined in 3.2.1, the system transient power sharing schemes are obtained by SVSDP and PSO-FS, respectively, and the impact results on gas turbine are analyzed and compared by Simulink simulation.

Rectangular and triangular LPPLs are selected as examples. Both LPPLs have consistent single-cycle energy with 6 operating cycles. The rectangular PPL features 0.1 s of power rising and falling edge time, 1.0 of relative peak power, with the power curve shown in Fig. 4.1. The

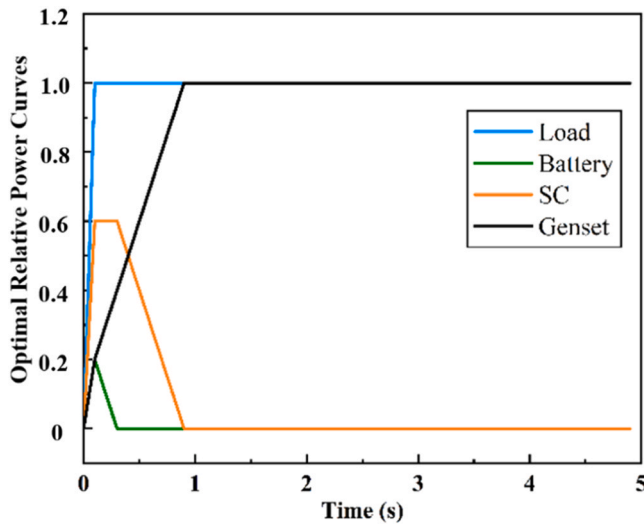


Fig. 3.8. System relative transient power sharing curves with $\Delta T_1=0.1$ s, $\Delta T_2=0.1$ s.

Table 3.5
PSO algorithm parameters.

Parameters	Values
Population size	80
Maximum iterations	30
Inertia coefficient	0.6
Learning factors	0.5/0.5
Search range for K_{SC}	[0.1,1]
Search range for K_B	[0.1,0.5]
Search range for T_{SC}	[0.1,1]
Search range for T_B	[0.5,2]

Table 3.6
Result of 5 groups of PSO fitness values.

Group	1	2	3	4	5
Fitness	0.796	0.789	0.795	0.770	0.769

triangular PPL features power rising and falling edge times of 4.9 s and 0.1 s, and the relative peak power of 2.0, with the power curve shown in Fig. 4.2.

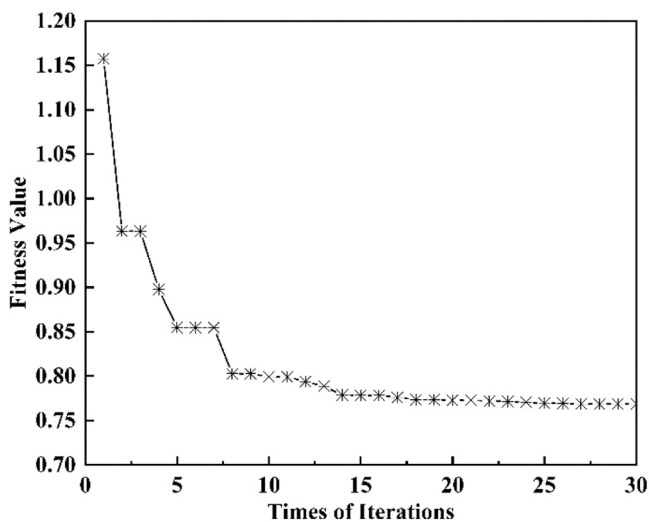


Fig. 3.9. Optimal personal fitness value.

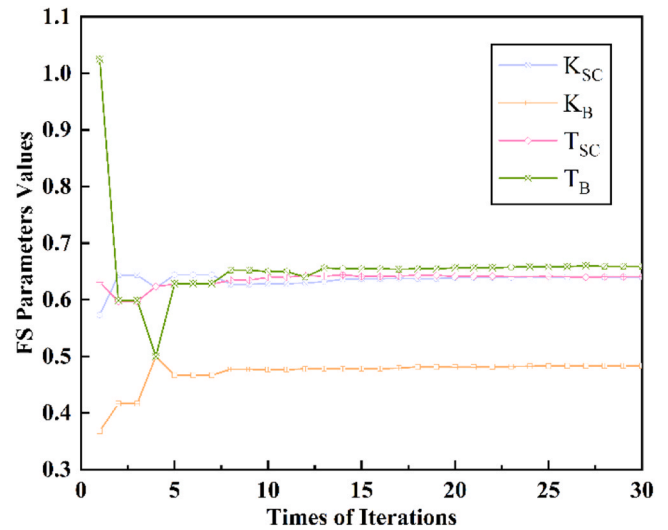


Fig. 3.10. Iterative process of FS parameters.

4.1. Transient power sharing for rectangular LPPL

4.1.1. Segmented variable step design

Referring to the exploration in 3.2.2, according to rectangular LPPL characteristic, the overall DP period is appropriately designated as 0–35 s, including the 30 s for operation and the 5 s for recovery. Segmented durations are as followed: $T_0=0.1$ s, $T_1=2 \times E_{SC} \cdot (SOC_{SCset} - SOC_{SCmin}) / P_{SCmax} = 1.2$ s, $T_2=5$ s, $2T_0-2T_1=2.4$ s, and the rational discrete step group are adopted: $[\Delta T_0, \Delta T_1, \Delta T_2] = [0.1, 0.2, 0.6]$. The DP segmented variable step design for rectangular LPPL can be derived as shown in Fig. 4.3.

4.1.2. SVSDP scheme for rectangular LPPL

Based on segmented variable step design in 4.1.1, the optimal objective value is obtained as 1.58 by SVSDP for rectangular LPPL. For comparison, the objective value obtained by PSO-FS is 4.56. Figs. 4.4–4.7 depict the parameters curves of the two schemes based on SVSDP and PSO-FS.

The transient processes of the two schemes are analyzed and compared. Within the first second (0–1 s), the output power of HESS in the SVSDP scheme is found to be higher, resulting in a reduction of the genset power ramp rate. At the moment of 5 s, 10 s, 15 s, 20 s, 25 s load

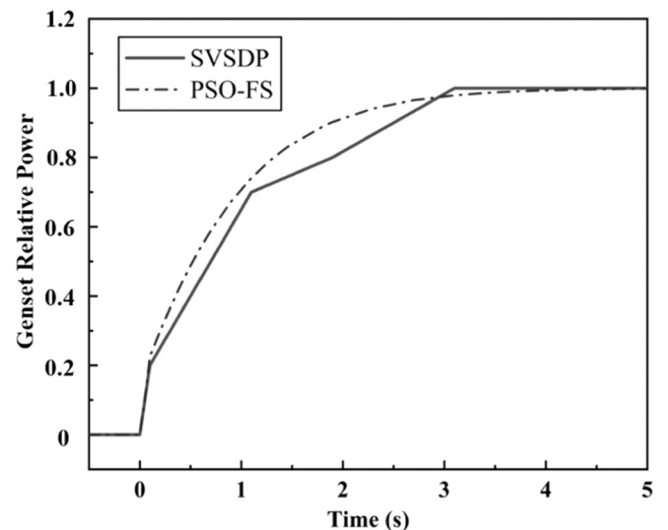


Fig. 3.11. Genset transient relative power of SVSDP and PSO-FS schemes.

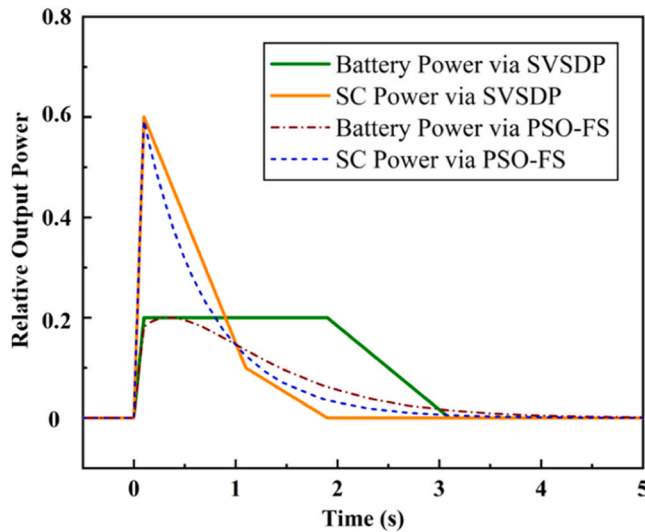


Fig. 3.12. HESS transient relative power of SVSDP and PSO-FS schemes.

repeated mutations, the HESS in SVSDP scheme demonstrates a capability to achieve pre-action. The compensations of battery energy and supercapacitor power are notably larger, effectively alleviating the genset power from mutations. In the recovery period of 30–35 s, the HESS input power is greater, and the battery performs reverse power output 2 s in advance, leading to an earlier power drop of the genset, significantly reducing the power ramp at the moment of load switch-out. From Fig. 4.7, it can be observed that both schemes maintain the HESS SOC within acceptable limits. However, in SVSDP scheme, the SOC of the batteries and supercapacitors exhibits significant variation in magnitude, suggesting a more comprehensive utilization of configured HESS.

4.1.3. Scheme analysis based on simulation

Utilizing a shipboard gas turbine model, the transient impacts on the genset in the SVSDP scheme are analyzed and compared with the PSO-FS scheme, and the simulation block diagram is shown in Fig. 4.8. The analysis is conducted through parameters such as power turbine speed, genset shaft stress, and gas generator turbine inlet temperature. (Liu et al., 2024) presented a detailed description of the modelling methods and validated the high accuracy against test data. In addition, the model

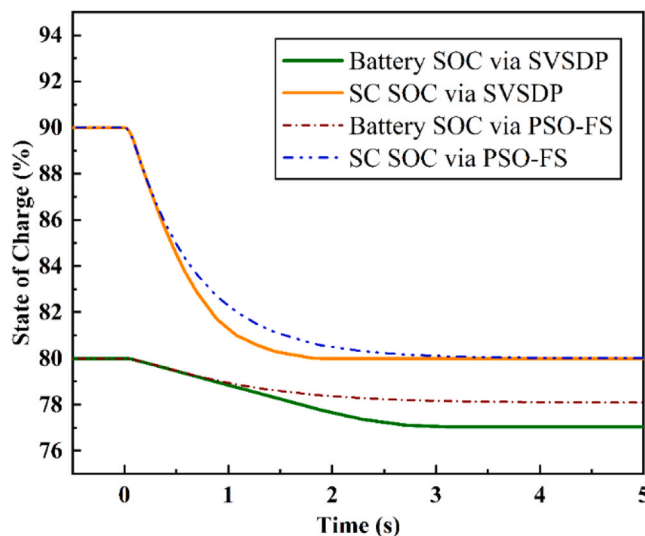


Fig. 3.13. HESS SOC of SVSDP and PSO-FS schemes.

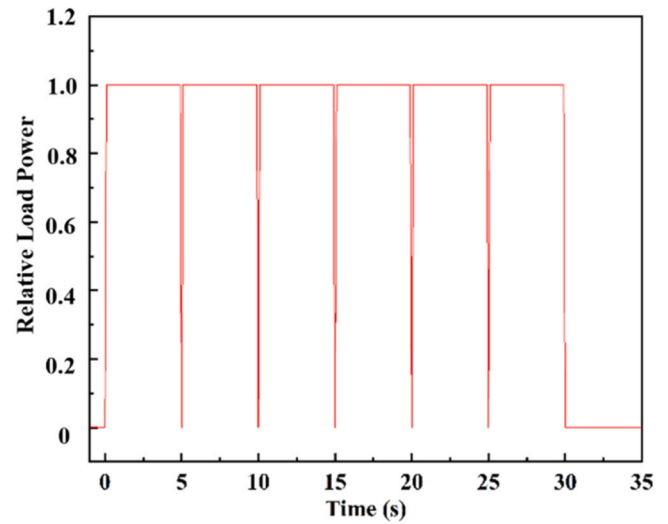


Fig. 4.1. Power curve of rectangular PPL.

has integrated the module for calculating shaft system with stress and fatigue damage. According to Fig. 4.4, it can be found that the rectangular LPPL could be triggered at genset 0–1.0 relative power conditions. Assuming that the rectangular LPPL is initiated at a genset relative power condition of 0.4, the resulting simulation results are shown in Figs. 4.9–4.11.

As can be seen from Fig. 4.9, comparing with PSO-FS, the SVSDP scheme decreases the TSR interval from $[-10.4\%, 8.3\%]$ to $[-9.9\%, 5.9\%]$, enhancing the transient stability of the rotor. From Fig. 4.10, it can be seen that in the SVSDP scheme, the stress alternation amplitude of the genset shaft is smaller, the stress transients of 5 s, 10 s, 15 s, 20 s and 25 s are weaker. The shaft fatigue damage of the process, calculated using the rain flow method (Lu et al., 2017), is determined to be 1.3943×10^{-9} , with a reduction of 53 % compared to PSO-FS scheme. The temperature curves in Fig. 4.11 reflect the internal state variation of the gas turbine. The maximum value of the gas generator turbine relative inlet temperature is 0.015 lower than that in the PSO-FS scheme. Additionally, the magnitude and ramp rate of the temperature variation are smaller, leading to a correspondingly reduced damage caused by thermal stress alternation.

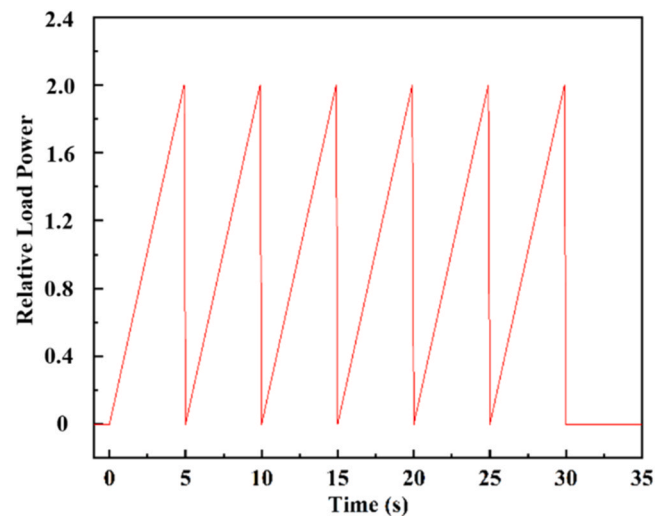


Fig. 4.2. Power curve of triangular PPL.

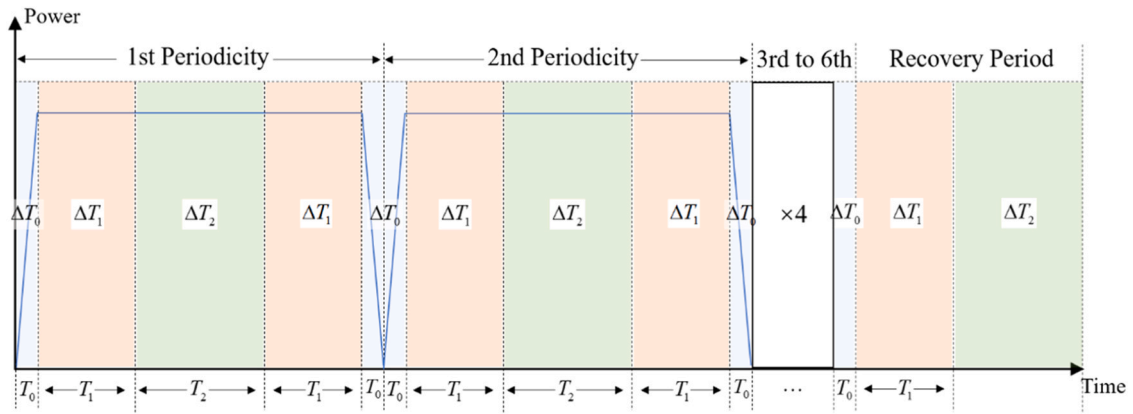


Fig. 4.3. DP segmented variable step design for rectangular LPPL.

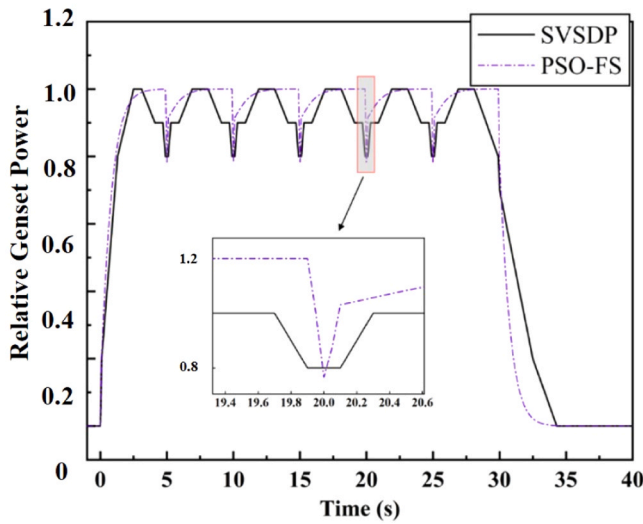


Fig. 4.4. Genset transient relative power of SVSDP and PSO-FS schemes.

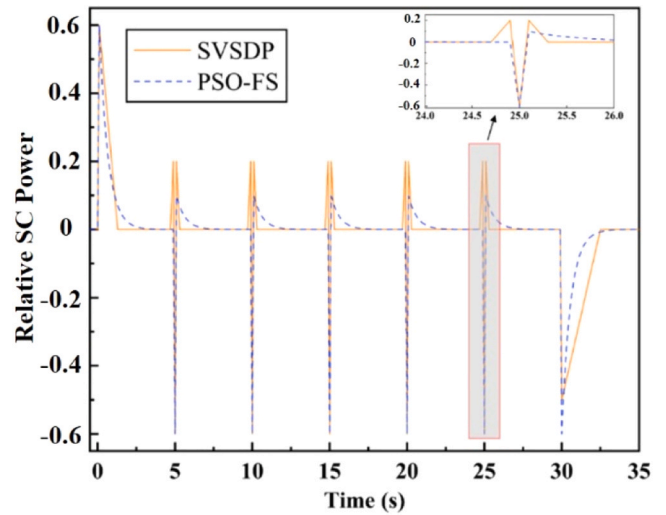


Fig. 4.6. SC transient relative power of SVSDP and PSO-FS schemes.

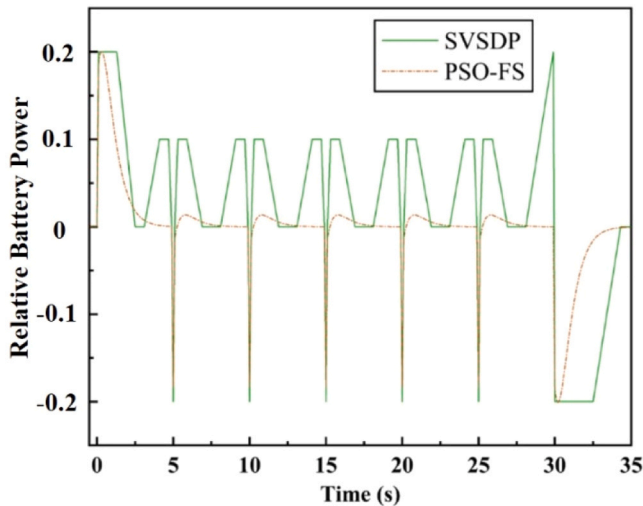


Fig. 4.5. Battery transient relative power of SVSDP and PSO-FS schemes.

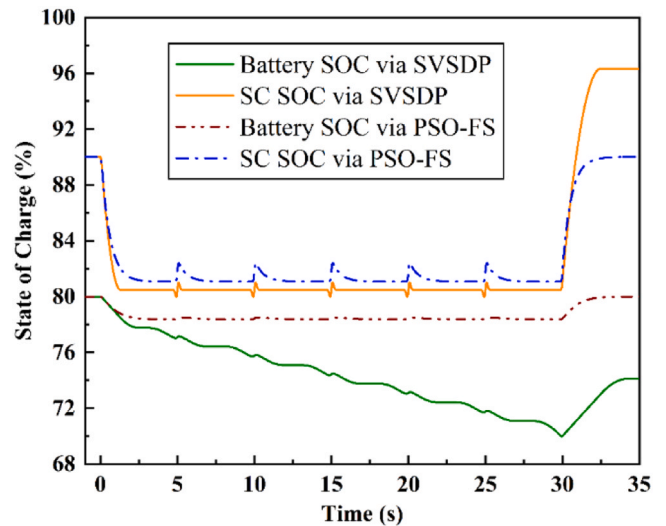


Fig. 4.7. HESS SOC of SVSDP and PSO-FS schemes.

4.2. Transient power sharing for triangular LPPL

4.2.1. Segmented variable step design

Similar to 3.2.2, according to triangular LPPL characteristic. Segmented durations are as followed: $T_0=0.1$ s, $T_1=1.2$ s, $T_2=2.4$ s, and

the rational discrete step group are adopted: $[\Delta T_0, \Delta T_1, \Delta T_2] = [0.1, 0.2, 0.6]$. The DP segmented variable step design for triangular LPPL can be derived as shown in Fig. 4.12.

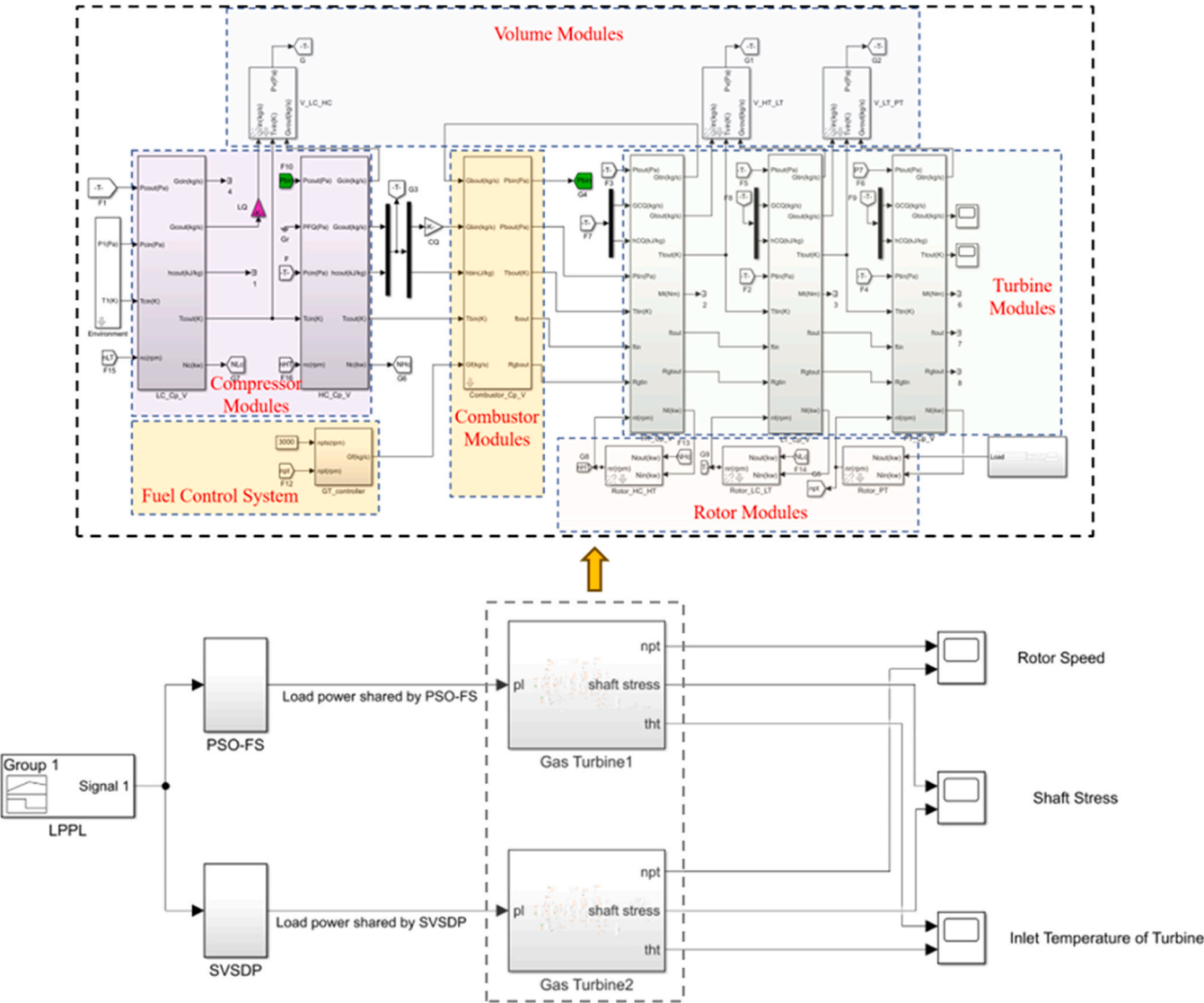


Fig. 4.8. Simulation block diagram for comparison of SVSDP and PSO-FS transient impact on genset.

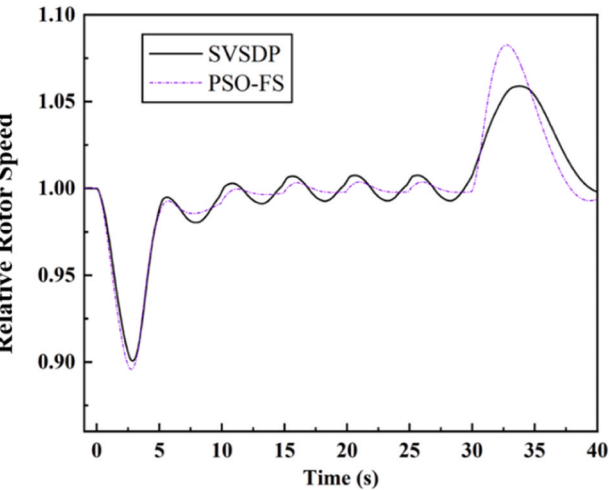


Fig. 4.9. Relative rotor speed with SVSDP and PSO-FS schemes.

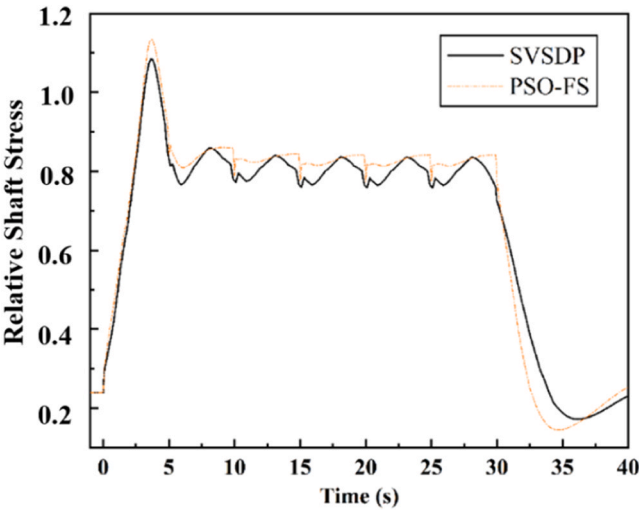


Fig. 4.10. Relative shaft stress with SVSDP and PSO-FS schemes.

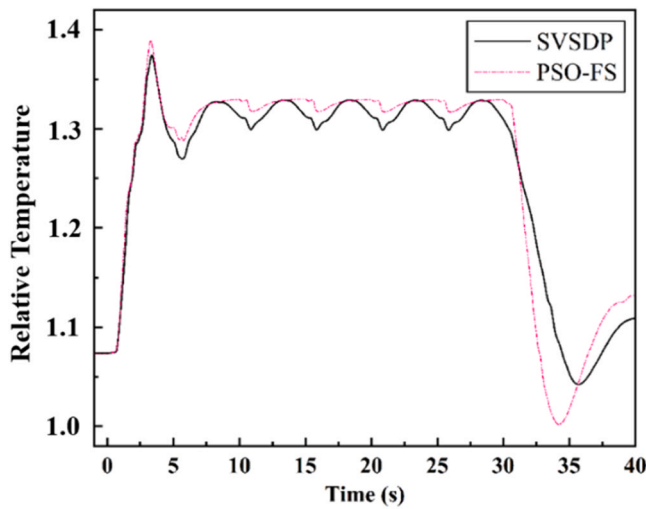


Fig. 4.11. Relative inlet temperature of gas generator turbine with SVSDP and PSO-FS schemes.

4.2.2. SVSDP scheme for triangular LPPL

Based on segmented variable step design in 4.2.1, the optimal objective value is obtained as 10.72 by SVSDP for triangular LPPL. For comparison, the objective value obtained by PSO-FS is 62.19. Figs. 4.13–4.16 depict the parameters curves of the two schemes based on SVSDP and PSO-FS.

The impacts of the triangular LPPL are concentrated in 6 power transient drops occurring at 5 s, 10 s, 15 s, 20 s, 25 s, 30 s, and the SVSDP scheme demonstrates greater specificity in suppressing the impacts. Before each abrupt drop in load power, the HESS increases the output power, reducing the magnitude of genset power transients. Fig. 4.15, illustrates that the supercapacitor precharges within 0–3 s. Although this slightly increases the genset power ramp rate, it ensures reaching a sufficiently high SOC at the 5 s to meet the discharge requirements, notably reflecting the global optimization capability of SVSDP. Additionally, Fig. 4.13 indicates that the amplitude of the genset relative power increases from SVSDP and PSO-FS scheme are approximately 1.2 and 1.8, respectively. Consequently, the allowed initial relative power range for the genset to operate the triangular LPPL is 0–0.8 for SVSDP and 0–0.2 for PSO-FS. This indicates that the SVSDP scheme broadens the starting conditions for the genset to operate triangular LPPL.

4.2.3. Scheme analysis based on simulation

Assuming that the initial relative power condition of the genset is

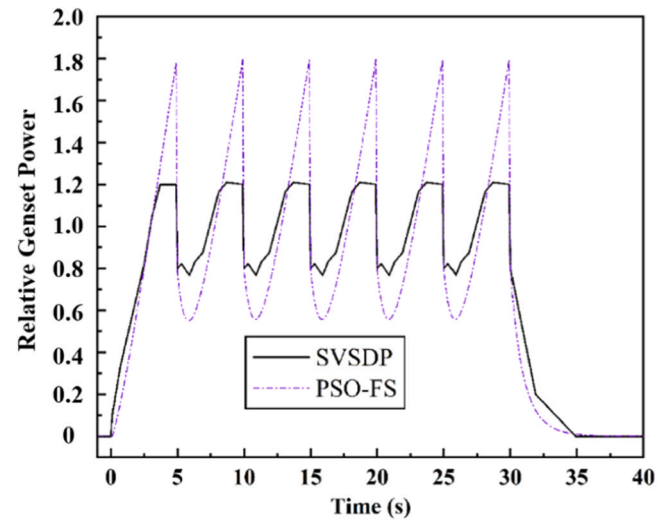


Fig. 4.13. Genset transient relative power of SVSDP and PSO-FS schemes.

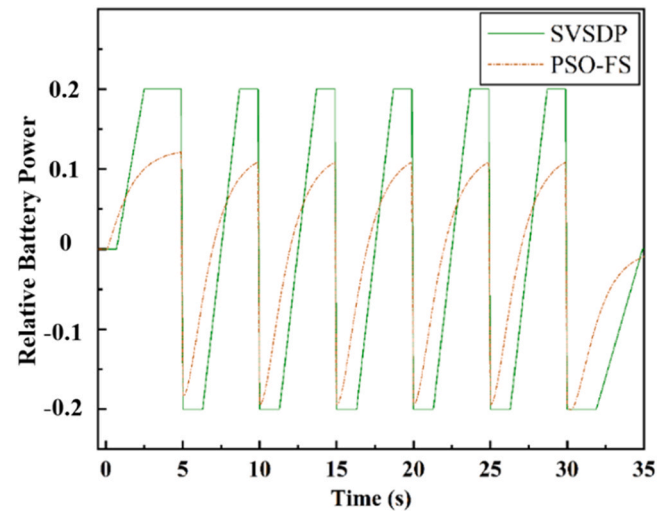


Fig. 4.14. Battery transient relative power of SVSDP and PSO-FS schemes.

0.2, before the triangular LPPL operation, the resulting simulation results are shown in Figs. 4.17–4.19.

As can be observed in Fig. 4.17, the PSO-FS scheme resulted in a TSR

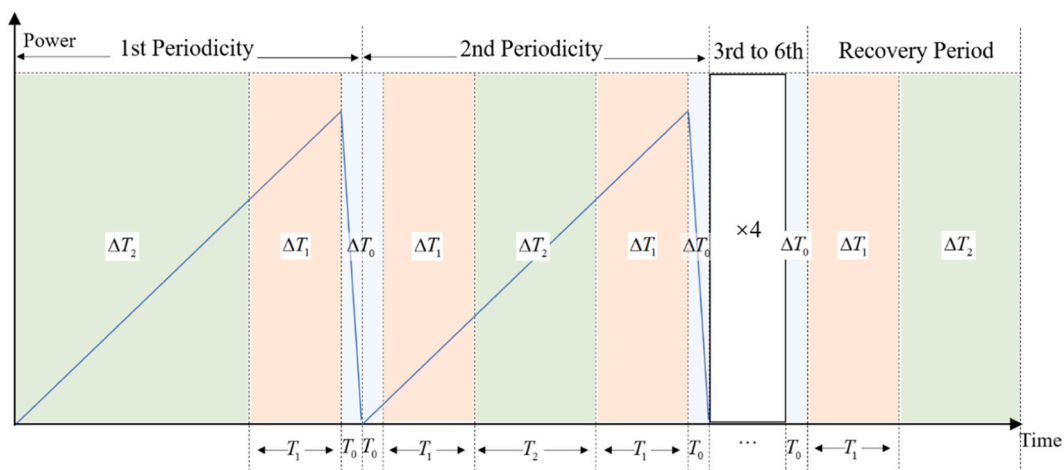


Fig. 4.12. DP segmented variable step design for triangular LPPL.

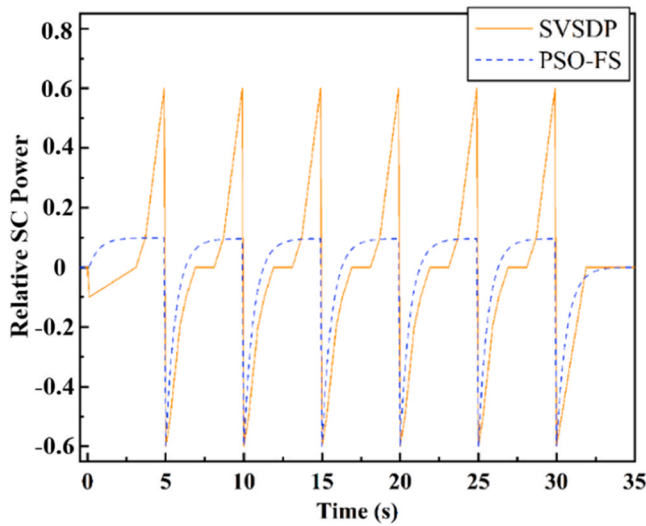


Fig. 4.15. SC transient relative power of SVSDP and PSO-FS schemes.

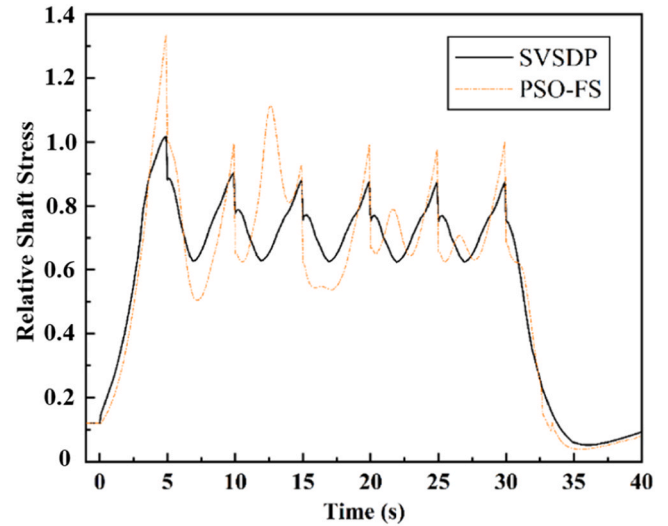


Fig. 4.18. Relative shaft stress with SVSDP and PSO-FS schemes.

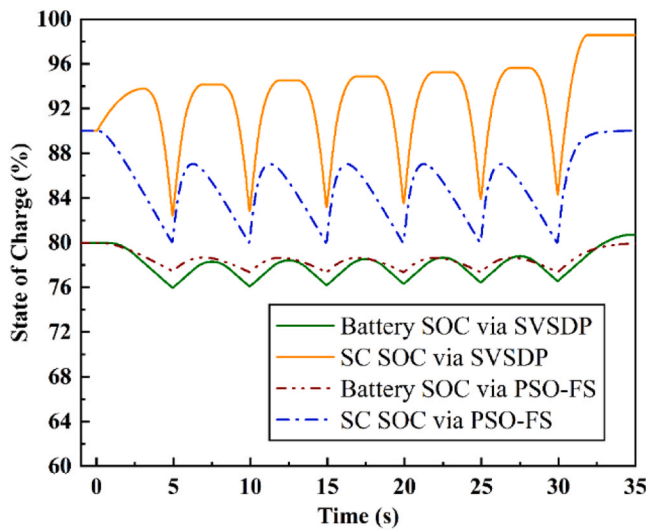


Fig. 4.16. HESS SOC of SVSDP and PSO-FS schemes.

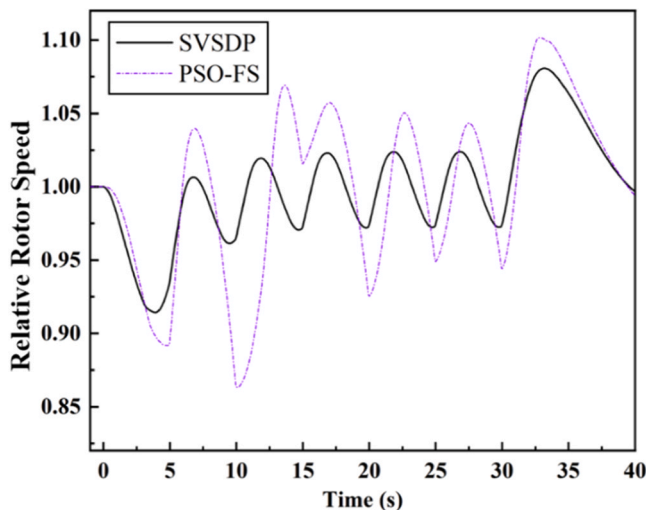


Fig. 4.17. Relative rotor speed with SVSDP and PSO-FS schemes.

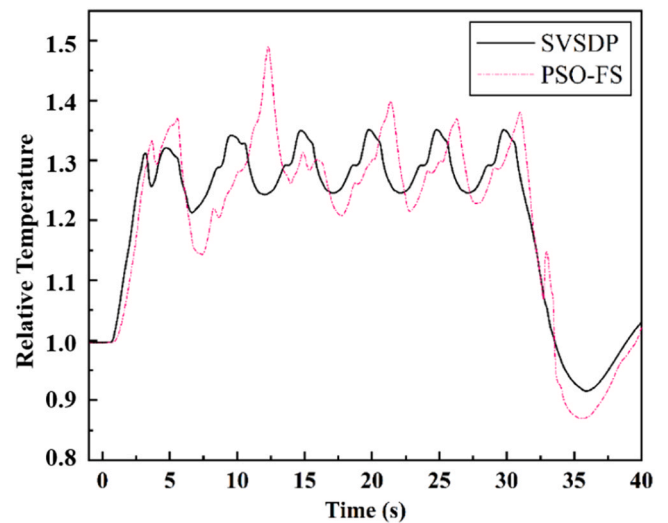


Fig. 4.19. Relative inlet temperature of gas generator turbine with SVSDP and PSO-FS schemes.

of $[-13.7\%, 10.2\%]$, which exceeds the $\pm 10\%$ limit, while the SVSDP scheme narrows the TSR to $[-8.6\%, 8.1\%]$. From Fig. 4.18, it is evident that in the SVSDP scheme, the stress alternation amplitude of the genset shaft is smaller. The fatigue damage to the shaft is calculated to be 9.8751×10^{-10} , indicating a reduction of 94.4 % compared to the 1.7705×10^{-8} of PSO-FS scheme. From Fig. 4.19, the magnitude and ramp rate of the temperature variation are smaller in SVSDP scheme, and the maximum value of the gas generator turbine relative inlet temperature is 0.139 lower than that in the PSO-FS scheme.

5. Conclusions

This paper proposes a segmented variable step dynamic programming method based on a shipboard MVDC system control architecture, which can be used to optimize system transient power sharing strategy in response to LPPL operation. The main contributions are as follows.

- (1) A shipboard MVDC system control architecture is proposed to achieve multi-sources coordinated control during LPPL operation.

- (2) Based on the dynamic programming theory, a transient objective function is established, and a segmented variable step-size dynamic programming method is proposed. This approach enables the DP theory to be applied to transient power sharing optimization in multi-sources hybrid system, reducing the transient impact on gas turbine genset.
- (3) In response to typical rectangular and triangular LPPL scenarios, the proposed method is employed for system transient power sharing, and system operation simulations are conducted. From the analysis of the proposed schemes and simulation results, it is evident that compared to traditional PSO-FS, the SVSDP scheme possesses the advantage of comprehensive global planning, allowing for a more effective utilization of HESS compensating capabilities during rapid fluctuations in electrical power. Under rectangular LPPL conditions, the genset TSR is reduced from 10.4 % to 9.9 %, the shaft fatigue damage is reduced by 53 %, and the maximum inlet relative temperature of gas generator turbine is reduced by 0.015. For triangular LPPL, the TSR is reduced from 13.7 % to 8.6 %, the shaft fatigue damage is reduced by 94.4 %, the maximum inlet relative temperature of gas generator turbine is reduced by 0.139, and the power range within which the system can operate triangular LPPL is increased in quadruple.

CRedit authorship contribution statement

Zhimeng Liu: Writing – original draft, Methodology. **Yongbao Liu:** Writing – original draft, Investigation. **Youhong Yu:** Writing – review & editing, Validation. **Rui Yang:** Writing – review & editing, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper

Data Availability

The data that has been used is confidential.

Acknowledgement

This paper is supported by the Basic Research for National Science and Technology Major Project of China (Grant No. J2019- $\times$ -0012) and the independent scientific research project of Naval University of Engineering (2023DY010).

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